

Dean Francisco Cabanes

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About Me

Economics student at Santa Barbara City College building Python tools for portfolio risk analysis, options pricing, and market modeling. Created DeanOS, a quantitative analytics engine with an interactive dashboard and AI agent interface. Seeking opportunities in finance and quantitative analysis.

Education

Santa Barbara City College | Economics | Santa Barbara, CA | 2026 - Present

- Current GPA: 4.0 | Coursework: Calculus, Microeconomics, Statistics, Public Speaking

Mira Costa High School | Manhattan Beach, CA | Graduated 2026

Technical Projects

DeanOS Portfolio Risk & Modeling Engine | Python, Next.js

- Built a portfolio analytics library with HMM/GMM market regime detection, GARCH volatility modeling, and 5,000-path Monte Carlo portfolio projections.
- Implemented three value-at-risk methods, OLS factor-exposure regression, historical and custom-shock stress tests, and hypothetical portfolio scenario comparisons.
- Connected the engine to an interactive Next.js dashboard and a Telegram AI agent for natural-language queries.

Non-Linear Beta Tracker | Python, FastAPI, React

- Built OLS and quadratic regression models with rolling, market-quantile, and volatility-regime betas to examine how equity sensitivity changes across market conditions.

Monte Carlo Options Pricer | Python

- Developed European, Asian, and barrier option pricing with antithetic variance reduction, standard errors, and 95% confidence intervals.
- Validated a 100,000-path NVDA simulation against Black-Scholes, with a price difference of 0.16%.

Experience

Spyder Surfboards | Hermosa Beach, CA | 2024 - Present

Assistant Director of Marketing | September 2025 - Present

- Manage Instagram and Facebook content end to end, including photography, video, product launches, and event promotion; continue supporting the business remotely while attending college.
- Generated 250,000+ views and 1,000+ shares, with approximately 200% engagement growth; promoted Nike SB releases that consistently sold out.

Sales Associate | May 2024 - Present

- Advise customers on surf gear and apparel, process transactions, and maintain inventory and displays.

Leadership & Activities

Los Angeles County Junior Lifeguards | Participant 2018 - 2023; Cadet 2024

Completed 130+ volunteer hours supporting ocean safety and youth programs.

Jimmy Miller Memorial Foundation | Surf Instructor | 2023 - 2024

Supported ocean therapy programs and helped participants develop surf skills safely.

Skills

- Python, scikit-learn, FastAPI, Next.js, React; regression, Monte Carlo simulation, risk modeling
- AI-assisted development with Claude and OpenClaw; communication, customer service, content creation

Interests

Surfing • Automotive projects • Camping • Travel • Financial markets